

UNIVERSIDAD TECNOLÓGICA DE TEHUACÁN

Nombre del alumno:

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8A

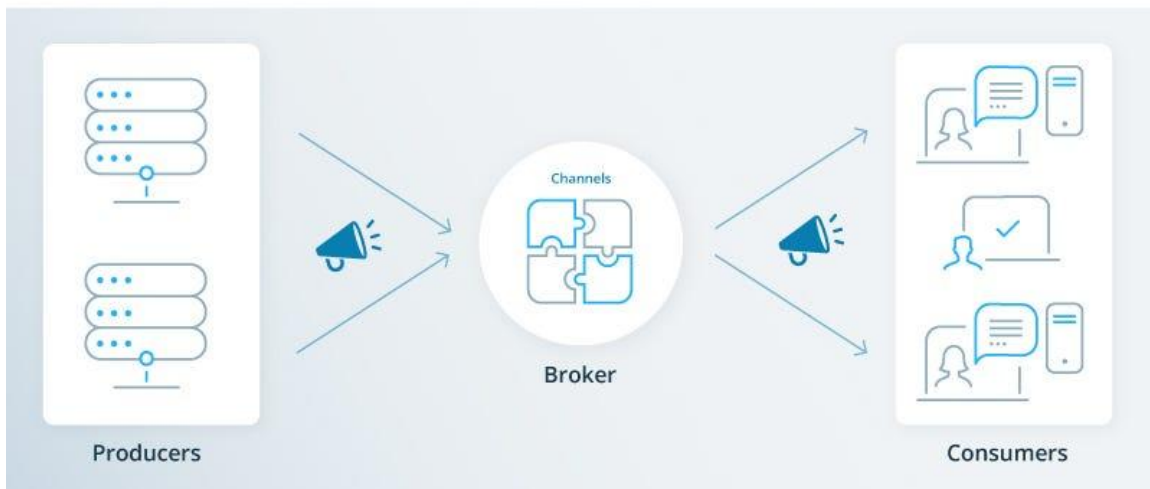
TSU EN TECNOLOGIAS DE LA
INFORMACION: DESARROLLO DE
SOFTWARE Y MULTIPLATAFORMA

TEHUACÁN, PUEBLA, 15 de febrero de 2026

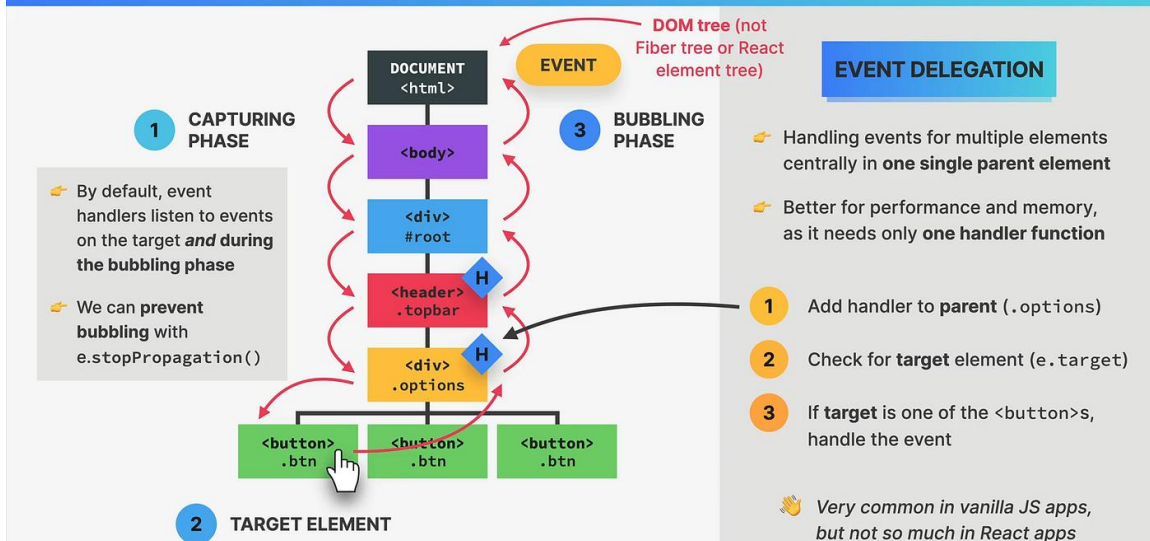
Eventos, estados y control por teclado

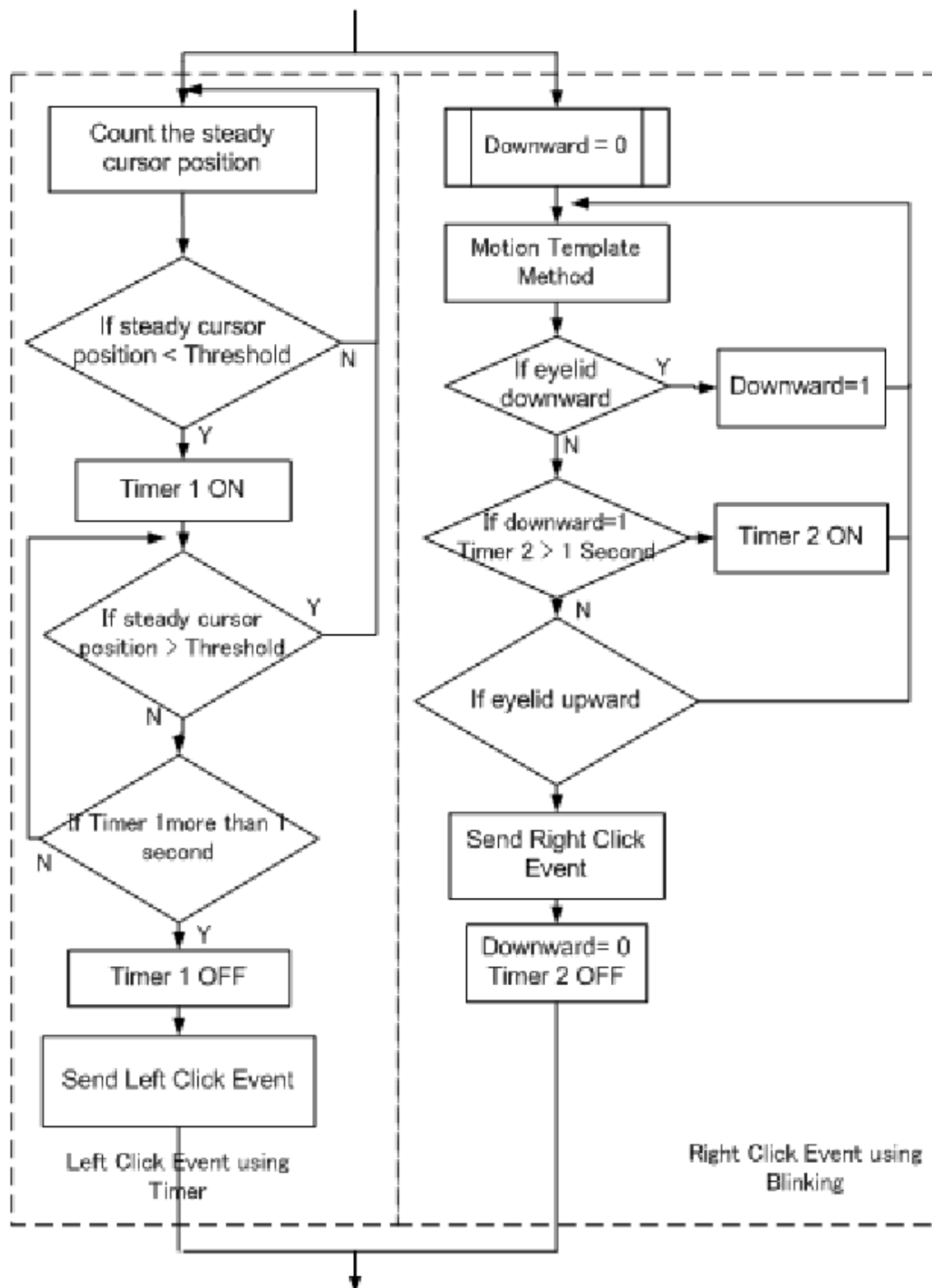
1 Event-Driven UI (Interfaz basada en eventos)

Event-Driven Architecture



DOM REFRESHER: EVENT PROPAGATION AND DELEGATION





Event-Driven Architecture

```
graph LR; ES(Event Source) -- "Button click event" --> EL(Event Listener); EL -- "subscribes to click event" --> EQ(Event Queue); EQ -- "stores events for processing" --> EL; EQ -- "checks queue for events" --> EV(Event Loop); EV -- "callback function" --> EH(Event Handler); EH --> EV;
```

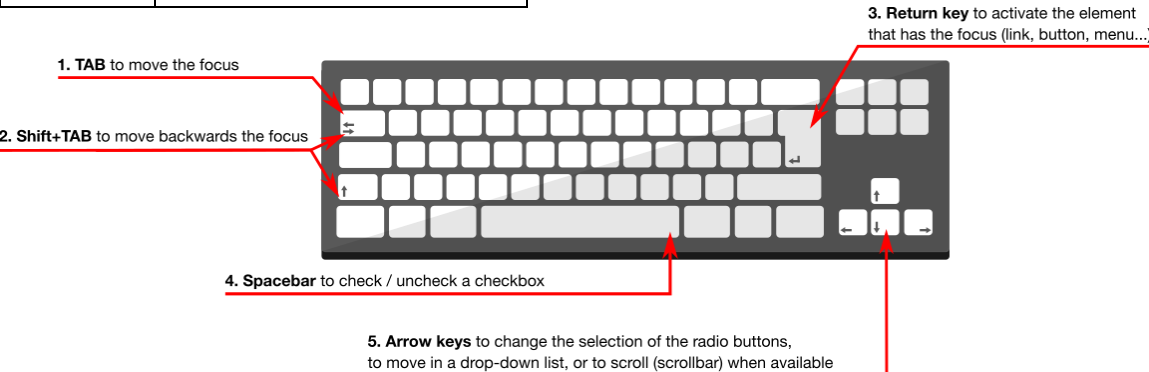
The diagram illustrates the Event-Driven Architecture (EDA) flow:

- Event Source**: Generates an event (e.g., "Button click event").
- Event Listener**: Subscribes to the event and passes it to the Event Queue.
- Event Queue**: Stores events for processing. It contains two "Event" objects.
- Event Loop**: Checks the queue for events and processes them.
- Event Handler**: Executes the callback function for the processed event.

Una **Event-Driven UI** es una interfaz que **no hace nada sola**, solo **reacciona** cuando el usuario provoca algo.

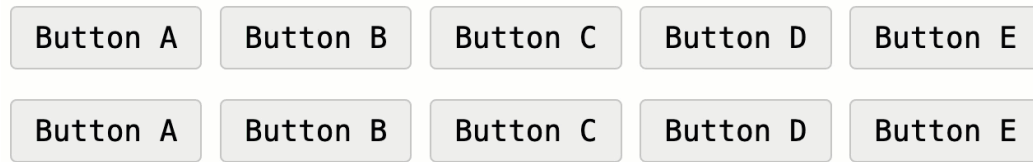
Ejemplos de eventos:

Evento	Qué significa
click	El usuario presionó algo
keydown	Presionó una tecla
focus	Entró a un campo
blur	Salió de un campo
submit	Envió un formulario



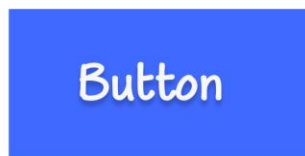
Using Both Tab And Arrow Keys Fo

Tab key moves focus from row-to-row. Arrow keys move focus from

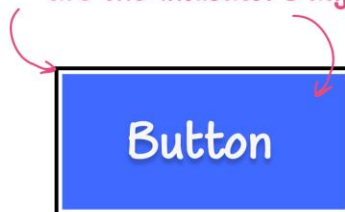


```
...  ▾ Elements Console Network Sources Perform
    <p x-data="tabGroup" @keydown.arrow-left="moveToPrevBut
      xToButton( $event )"> == $0
      <!--
        The first button has a tabIndex of "0" so
        key. All other keys in this group can be
        -->
        <button tabindex="0"> Button A </button>
        <button tabindex="-1"> Button B </button>
        <button tabindex="-1"> Button C </button>
        <button tabindex="-1"> Button D </button>
        <button tabindex="-1"> Button E </button>
      </p>
    > <p x-data="tabGroup" @keydown.arrow-left="moveToPrevBut
```

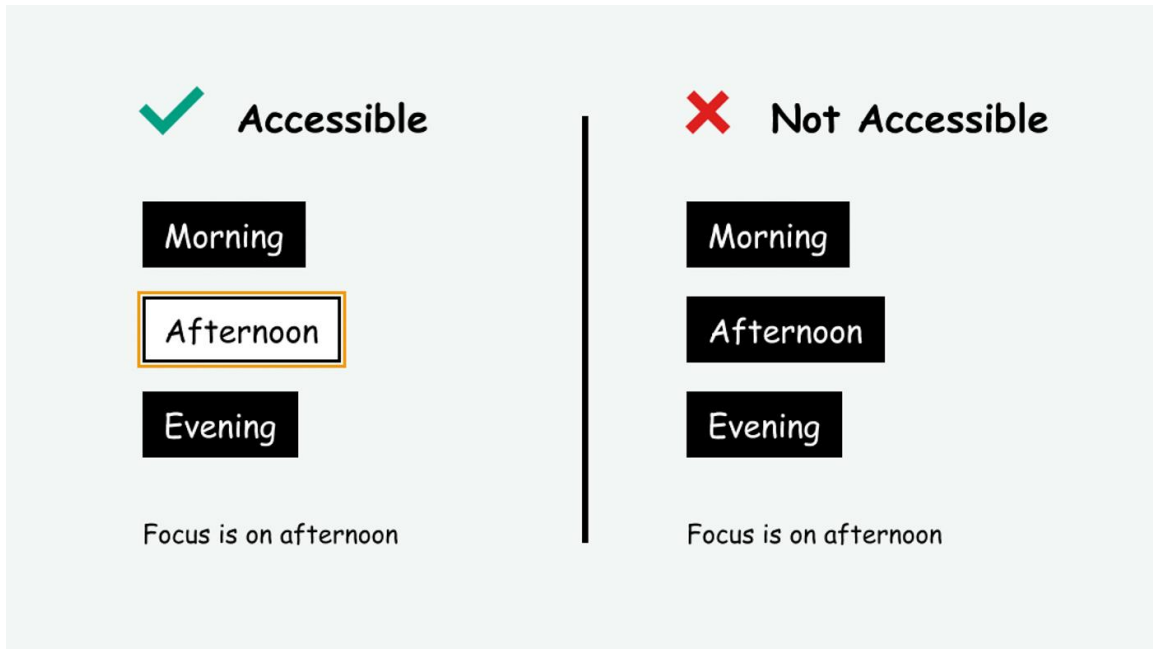
The button's border and background colors are the indicator's adjacent colors.



The button, unfocused.



The button, focused.



La regla es:

Nada debe depender solo del mouse.

Eso significa:

- Tab → moverte entre elementos
- Enter → activar botones
- Space → seleccionar
- Escape → cerrar
- Flechas → navegar

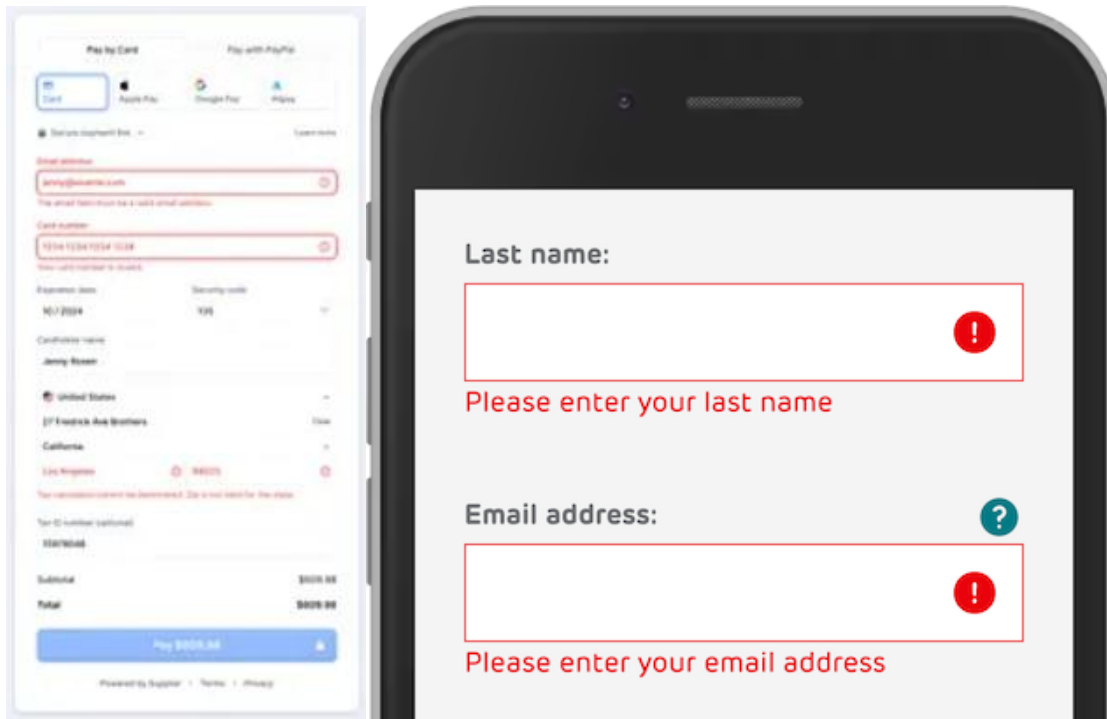
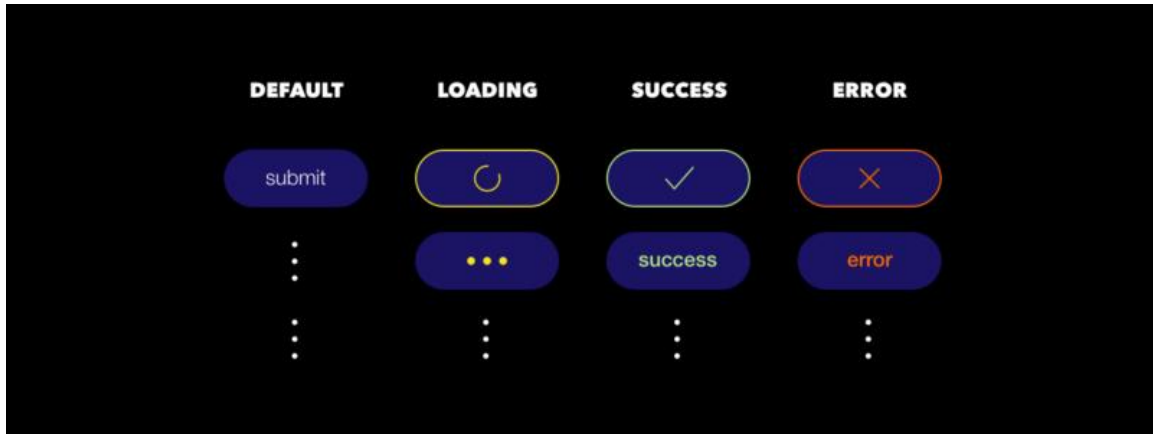
Ejemplo real:

Si tienes un botón “**Guardar**”, debe funcionar:

- con clic
- con Enter
- con Space

Si no → **no es accesible** ❌

3 Estados visibles y accesibles





Message at top of the form informing user of issues

❗ **There was a problem with your submission. Please review the fields below.**

Your Name *

Please enter your full name

Please enter your full name

This field is required. Please fill in your full name.

Your Email Address *

chandesh"nerodigital.co.uk

Please enter your email address

This field is required. Please fill in your email address.

Change of background colour and field outline help to quickly identify errors

Error message is close to the field

Un **estado** es la situación en la que se encuentra tu interfaz.

Ejemplos:

Estado	Significa
idle	Esperando
loading	Procesando
success	Se logró
error	Falló
disabled	No disponible
focused	Seleccionado

Y deben ser:

- 👁 **Visibles** (colores, iconos, mensajes)
- 🗣 **Accesibles** (lectores de pantalla los leen)

Ejemplo profesional:

Si un login falla, no basta con “no entrar”.

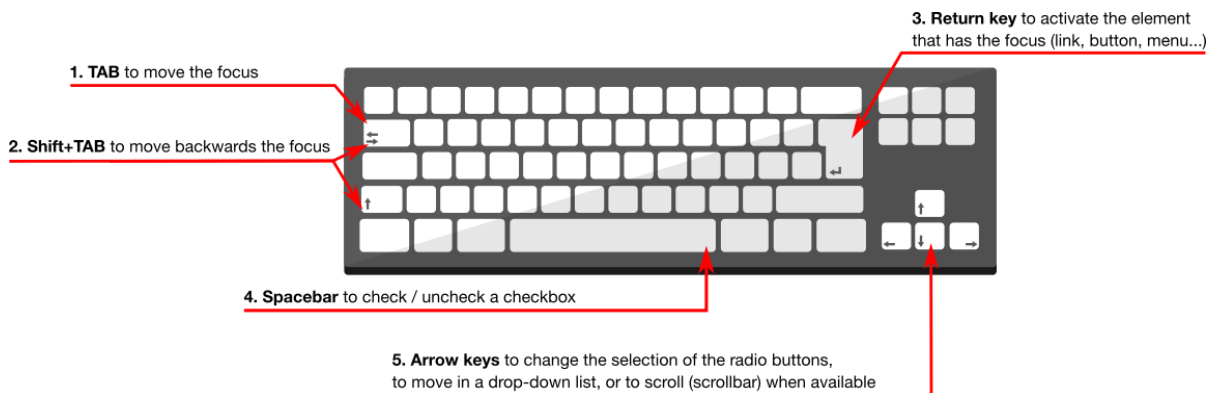
Debe decir:

“Contraseña incorrecta”

Eso es **feedback accesible**.

Requisito	Significa
Eventos mouse + teclado	Todo funciona con clic y con teclas
Estados claros	Se sabe cuándo algo carga, falla o está listo
Feedback accesible	El sistema “habla” con el usuario
Tests	Demuestras que funciona

¿Qué son los tests de interacción?



UI Testing vs UX Testing - What's the Difference?

Aspect	UI Testing	UX Testing
Focus	Focuses on the interface's look and feel.	Focuses on the overall user experience.
Scope	Limited to interface elements like buttons, colors, typography, and layouts.	Encompasses usability, accessibility, and user satisfaction.
Methodology	Checks visual consistency, alignment, and responsiveness.	Involves user surveys, A/B testing, and observing user interactions with the product.
End Goal	Ensures interface functionality and visual appeal.	Ensures product intuitiveness, usability, and meets user needs.
User Involvement	Does not require direct user involvement.	Relies heavily on user feedback and interaction for insights into user experience and needs.



Accessible

1 Street address
123 Square Street

2 City
Square City

3 State
SQ ✓

4 Zip code
2416256



Not Accessible

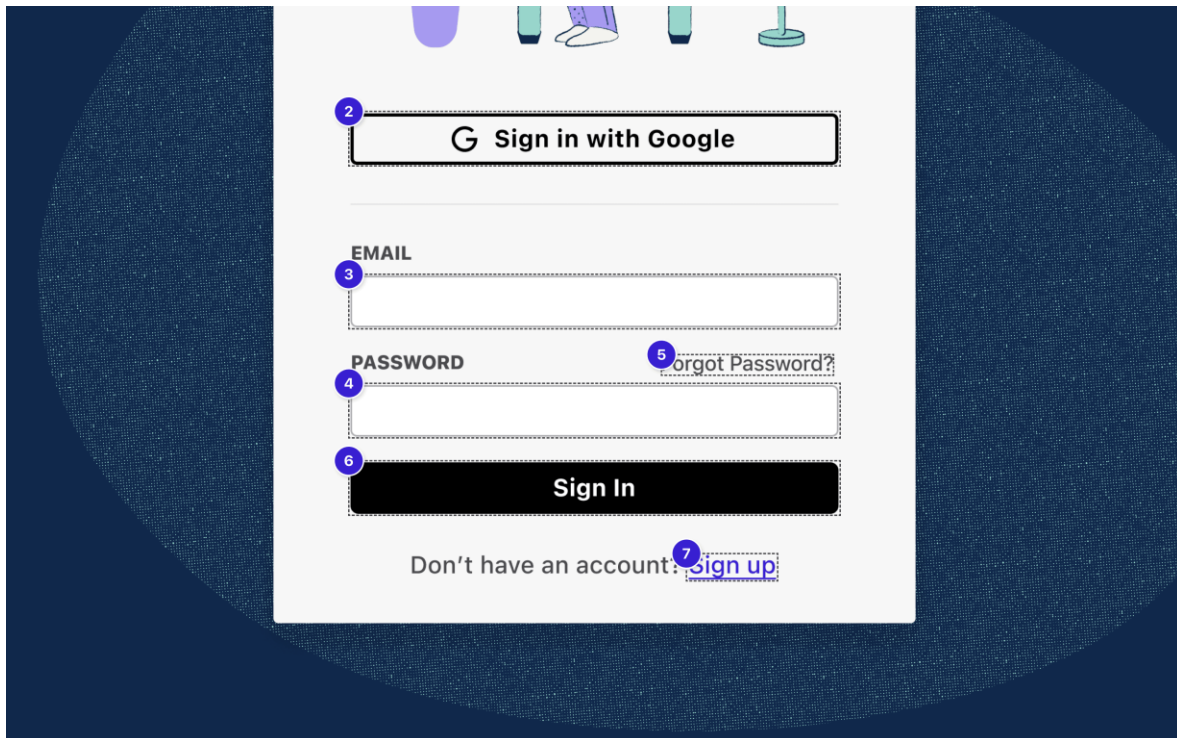
1 Street address
123 Square Street

3 City
Square City

4 State
SQ ✓

2 Zip code
2416256

(Numbers = focus order)



No son pruebas automáticas (aún).
Son casos de prueba como estos:

Acción	Resultado esperado
Presionar TAB	El foco pasa al siguiente botón
Presionar ENTER	Se activa el botón
Error en formulario	Se muestra mensaje accesible
Campo inválido	Se marca visualmente

Esto demuestra que tu interfaz sí obedece eventos.